Calibo AI Academy · Trainer Reference · Not for student distribution

MUC02 — Trainer Notes

Credit Pre-Screening Engine — Loan Intake Optimisation · Path 2, KLU Batch

Phase 2 — Supervised Learning Part 1 | BFSI — Retail Lending

> **Do not share this document with students.** Distribute the student document and dataset CSV only.

1. Before the Session

Item	Detail
Share with students	MUC02_KLU_Credit_PreScreening.docx + MUC02_Credit_PreScreening_Dataset.csv
Do not share	This trainer notes document, or the expected-outcomes deck until submissions are in
Session duration	2.5 hours. Submission deadline: 48 hours after session.
Deadline	Add to the student document before distributing.
Environment check	Confirm students have pandas, numpy, matplotlib, seaborn, scikit-learn and jupyter installed. Run a 5-minute check at the start.
Curriculum position	Students have completed Phase 1 and Phase 2 Modules 1–2 (Supervised Learning Part 1, including logistic regression). Trees, ensembles, clustering and cross-validation have not been taught — do not expect them.

2. Dataset Notes

The dataset is the same file for both the KLU Path 2 and Path 3 batches. This is intentional — cross-batch consistency at the data and stakeholder level, as with MUC01. Both batches answer to the same Credit Policy Analyst. Path 3 students go further: they compare four models and design a two-lane pre-screener. KLU students stop at a single logistic regression and one auto-decline threshold.

12,000 applications, April–September 2026, 19 columns. Overall approval rate: 52.9%. The target is reasonably balanced, so accuracy is not misleading on its own — but students should still be pushed past accuracy towards the cost of wrong declines.

2.1 Planted data issues (Step 5.1)

Issue	What students should find and do
Missing monthly_income — 441 rows	Concentrated in Self-Employed Professional and Business Owner applicants (about 11–13% of each). Reasonable fix: fill with the median income for that employment type. Dropping rows is acceptable only if justified. Distinction-level: noticing that missing income is itself a self-employed pattern.
Blank credit_score — 1,039 rows	Not errors. These are new-to-credit (NTC) applicants — credit_history_months is 0 for all of them. They must be kept and flagged (is_ntc). Students who drop them have removed about 9% of applicants, many of them young.
Implausible monthly_income — 12 rows above ₹6,00,000	Annual income entered as monthly, all Salaried. Fix: divide by 12. Students who leave them in get a few absurdly low FOIR values; students who drop them are acceptable if stated.

2.2 Key patterns students must find

Pattern	Numbers (reference)	What a good student says
Credit score cliff at 650 (5.2)	600–649: 4.3% approved. 650–699: 32.8%. 750+: about 82%.	"The bank effectively has an unwritten floor at 650 — below it, applications are almost never approved. That is a rule a pre-screener can apply with confidence."
No-score applicants sit in the middle (5.2)	NTC: 53.9% approved.	"New-to-credit applicants are approved about half the time — the credit team is clearly using judgement here, so the model will find them hard."
FOIR cliff above 50–55% (5.3)	Below 30%: 65.4%. 50–55%: 31.7%. Above 55%: 15.5%.	"Once EMIs take more than about half of income, approvals collapse. I would propose a FOIR limit of 50–55% in policy."
Gig / Contract applicants approved least (5.2)	Gig: 34.9%. Salaried: 58.4%.	Linked to income stability and shorter job tenure — not just the label.
Raw gender gap that mostly disappears (5.6)	Male 54.1%, Female 50.4%. Within Gig and Business Owner the gap is under 1 point.	"The gap comes mostly from income and employment mix — female applicants have lower median income (₹45K vs ₹55K) and are twice as likely to be gig workers (21% vs 12%)."

3. Reference Results

> Student numbers will differ slightly depending on cleaning choices. Accept results within about ±2–3 percentage points. If a student is far outside this range, check the split (stratify, random_state=42), the target direction, and whether gender or decision leaked into the features.

3.1 Logistic regression (Step 5.4)

Metric	Value	Notes
Accuracy	79.1%	Test set of 2400 applications
ROC-AUC	0.867	
Precision (Declined)	79.1%	Of those predicted Declined, how many were actually declined
Recall (Declined)	75.5%	Of actual declines, how many the model caught
Confusion matrix	TN 1046 · FP 225 · FN 277 · TP 852	FP = 225 creditworthy applicants who would be wrongly declined at 0.5. This is the number to push students towards.
Model accuracy, scored vs NTC	81.2% vs 56.7%	The model is barely better than a coin toss on new-to-credit applicants.

Largest coefficients (scaled features; positive = pushes towards decline): FOIR (+1.22), credit score (−1.01), delinquencies in 24 months (+0.98), Gig / Contract (+0.64), Debt Consolidation purpose (+0.51), Salaried (−0.42), new-to-credit flag (+0.38).

3.2 Threshold table (Step 5.5)

Threshold	% auto-declined	% wrongly declined	Declines caught
0.50	44.9%	20.9%	75.5%
0.60	37.6%	16.8%	66.5%
0.70	30.8%	12%	57.6%
0.80	23.4%	7.1%	46.2%
0.85	19.7%	5.9%	39.3%
0.90	15.2%	4.7%	30.7%
0.95	10%	2.1%	20.8%

> **Expected answer: threshold 0.90.** It is the lowest threshold that keeps wrong declines under 5% (4.7%). It auto-declines 15.2% of applications — about 456 a month out of 3,000, or roughly 190 analyst hours a month. At 0.85 wrong declines are 5.9%, which breaks the rule. Accept 0.88–0.92 if the student's own table supports it.

The teaching point: at the default 0.5, the model would auto-decline 44.9% of applications but 20.9% of them would be people the bank actually approved. The bank sets the threshold from its risk appetite, not from the model.

3.3 Fairness (Step 5.6)

At the 0.80–0.90 range, the model's auto-decline rate is almost identical by gender (about 24.1% female vs 23.1% male at 0.80). The model does not widen the historical gap. Distinction-level students note that income and employment type can act as proxies, that leaving gender out does not by itself guarantee fairness, and that outcomes by gender should be monitored after launch.

4. Step-by-Step Coaching Notes

5.1 — Load, inspect and clean

> The most common failure: students drop every row with a null, losing all new-to-credit applicants without noticing. Ask: "Who are these people with no credit score? What does credit_history_months say about them?" The cleaning log in markdown matters as much as the code.

5.2 — What drives approval today?

> Push students to name the cliff and turn it into a policy statement. "Lower scores get fewer approvals" is a description. "Below 650 the bank almost never approves — a pre-screener can apply that rule safely" is an interpretation.

5.3 — Affordability features

> FOIR is the single most important feature in the dataset and students must build it themselves. If students are stuck, point them back to the definition box in Section 4 of their document. For the bands, hint: pd.cut(df["foir"], bins=[0,0.3,0.4,0.5,0.55,10]).

5.4 — Logistic regression

> Watch the target direction — the model predicts declines (Declined = 1). Students who flip it will read every coefficient backwards. Check that gender, city and decision are not in the feature list. Push for the confusion matrix in business language: "225 applicants the bank would have approved are wrongly declined at 0.5."

5.5 — Threshold

> This is the heart of the use case. Students who choose a threshold without the table, or without the 5% rule, have missed the point. The monthly translation (applications and analyst hours) is what makes this a business answer.

5.6 — Fairness

> Keep it short but real. The trap answer is "we did not use gender, so the model is fair." The better answer measures outcomes by gender, explains the gap through income and employment mix, and recognises proxy variables. Do not let this become a lecture — Responsible AI is covered properly in Phase 3.

5.7 — Credit Policy Briefing

> Read every briefing carefully. Pass: states a threshold. Distinction: states the threshold with applications and hours per month, names the two biggest drivers with numbers, names new-to-credit applicants as a group to keep with humans, and takes a clear fairness position.

5. CBIM Canvas — What to Look For

Field	Common weak answer	What to look for instead
Situation (S)	A bank gives loans	A mid-size bank receives ~3,000 personal loan applications a month and reviews every one manually, at ~25 minutes each

Field	Common weak answer	What to look for instead
Complication (C)	Too many applications	Analyst time is spent on clear-cut declines; good applicants wait longer; the bank cannot scale without hiring
Question (Q)	Build a model to predict loans	Which applications can be safely removed from manual review, and at what threshold, without wrongly declining more than 5% of creditworthy applicants?
SSOT	The dataset	MUC02_Credit_PreScreening_Dataset.csv — and the decision column is the credit team's past judgement, not whether the customer repaid
Success definition	High accuracy	A threshold that respects the 5% risk limit and frees up a stated number of analyst hours per month
Stakeholder	Credit Policy Analyst	Credit Policy Analyst — and the analysts whose work changes, the Head of Retail Lending, and the applicants affected

> **Worth raising in class:** the target is the analysts' historical decision, not loan repayment. The model learns to copy past decisions — including any bias in them. Students who spot this in the SSOT field are thinking at distinction level.

6. Grading Reference

Grade band	Characteristics
Distinction (85–100)	All seven steps complete. Cleaning decisions justified, NTC applicants kept. FOIR built and its cliff identified. Threshold chosen against the 5% rule with monthly hours stated. Fairness answer goes beyond "we did not use gender." Briefing is specific and quantified.
Pass (65–84)	All steps present, code runs. Model built correctly. Threshold chosen but impact not fully quantified. Markdown sometimes describes rather than interprets.
Marginal (50–64)	One or two steps incomplete. Threshold left at 0.5 or chosen without the table. NTC rows dropped without comment. Briefing vague.
Redo (below 50)	Notebook does not run end-to-end, or gender/decision leaked into features, or three or more steps missing. CBIM canvas absent.

7. Common Student Questions

Question	Suggested answer
"Can we drop the rows with missing values?"	Only if you can justify it. Blank credit scores are new-to-credit applicants — dropping them removes a real group of customers. Missing incomes are mostly self-employed — think about what dropping them does.
"Why are we predicting Declined instead of Approved?"	The use case is about auto-declining safely, so the class we care about is Declined. Either direction works mathematically, but be consistent.
"Is 79% accuracy good enough?"	Accuracy is the wrong question. Ask how many good customers get wrongly declined at your threshold, and whether that is within the bank's limit.
"Can we use gender to improve the model?"	No. Using gender in a credit decision is not something a bank could defend. Measure outcomes by gender instead.
"Can we use a different threshold list?"	Yes, as long as it covers 0.5 to 0.95 and identifies where the 5% rule is met.

For questions not covered here, use your judgement and align with the programme lead if needed.